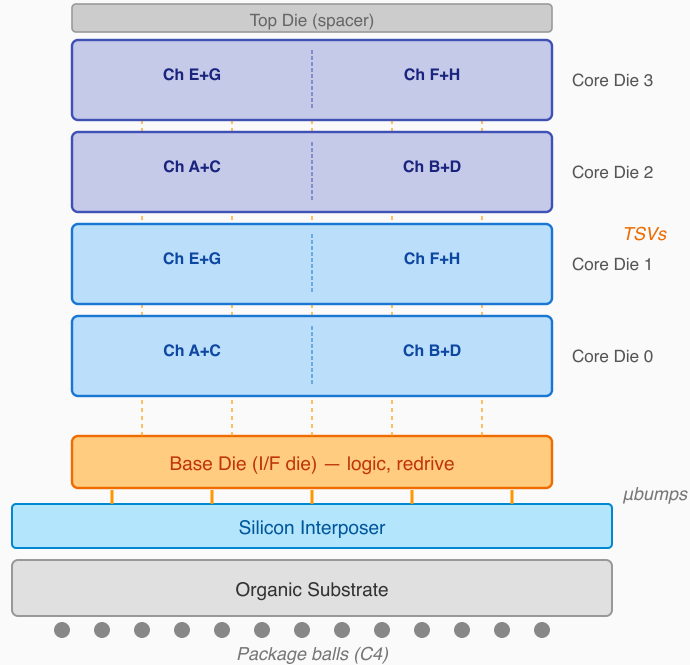


HBM2E Stack Cross-Section: 4-High vs. 8-High Configuration

Channel-to-die mapping shown; both configurations have 8 channels (A–H) and identical bandwidth per stack

4-High (e.g. A100-40GB HBM2)

4 core dies + 1 base die = 8 GB/stack



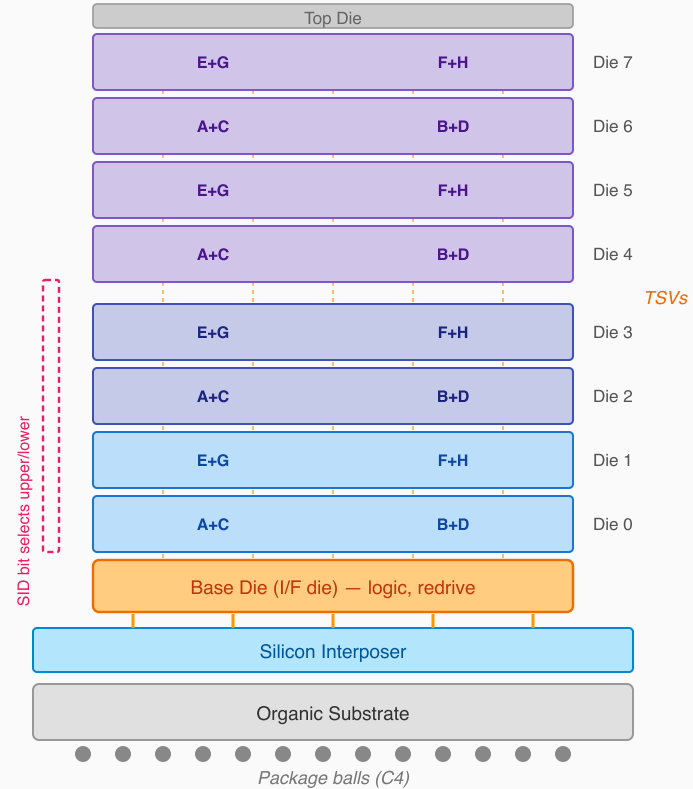
4 dies × 2 GB/die = 8 GB per stack

5 stacks on A100-40GB → 40 GB total

Max ~2 Gbps/pin → 256 GB/s per stack (HBM2)

8-High (e.g. A100-80GB HBM2e)

8 core dies + 1 base die = 16 GB/stack



8 dies × 2 GB/die = 16 GB per stack

5 stacks on A100-80GB → 80 GB total

Max 3.2 Gbps/pin → 410 GB/s per stack (HBM2e)

Both configurations: 8 channels × 2 pseudo channels = 16 independent access points per stack.

8-High doubles capacity via SID (stack ID) addressing bit; bandwidth increase is from higher pin rate, not more channels.